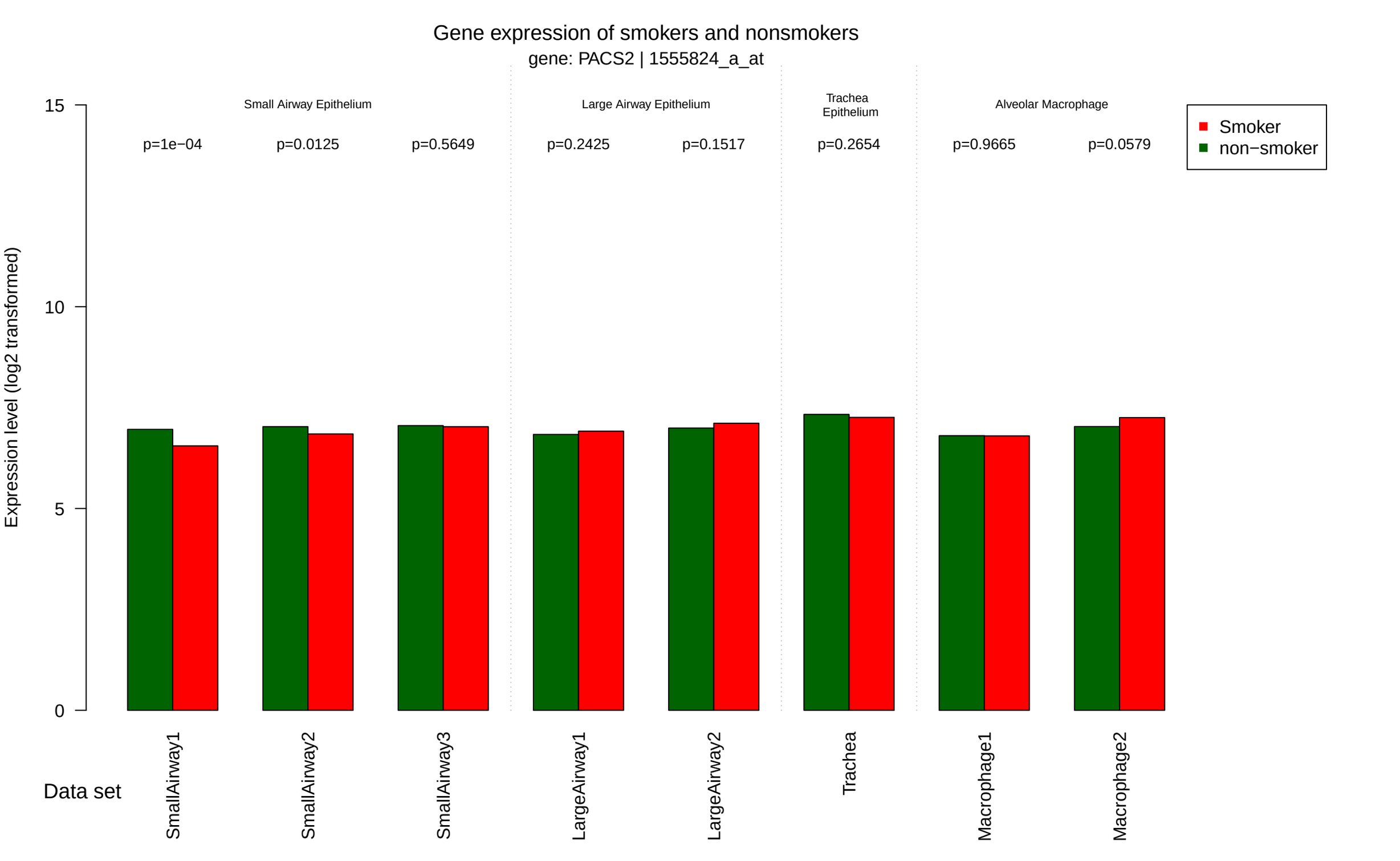
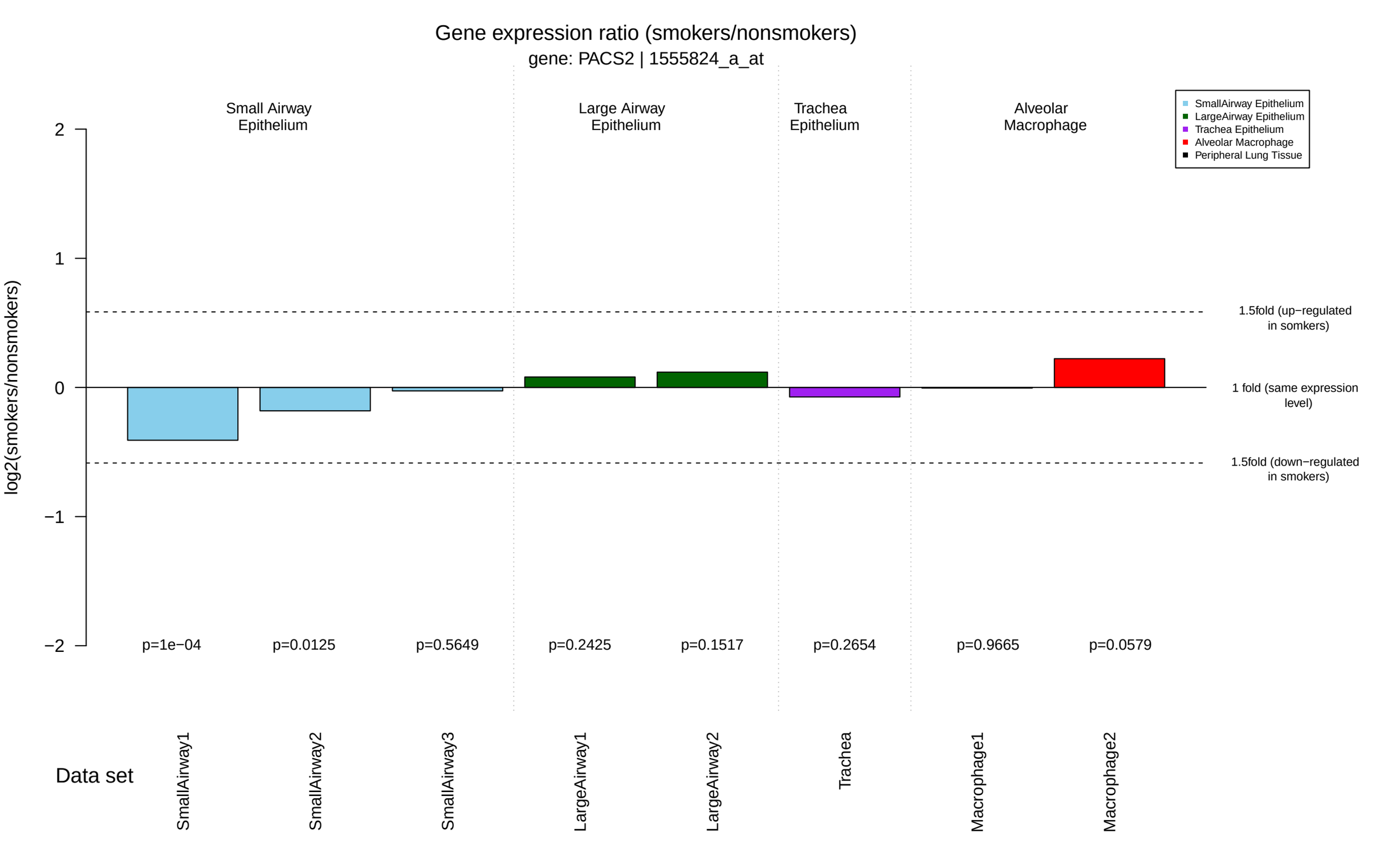
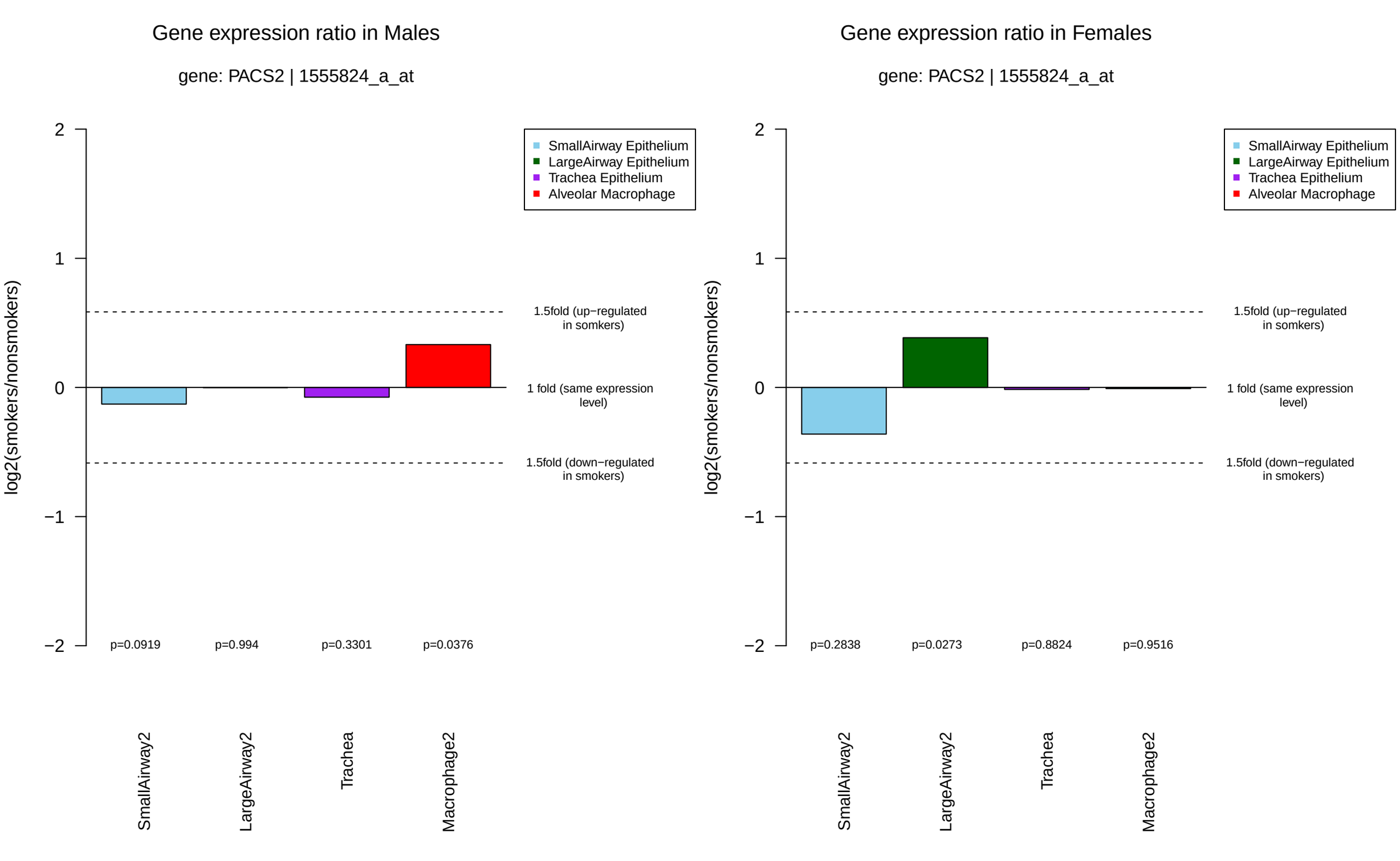
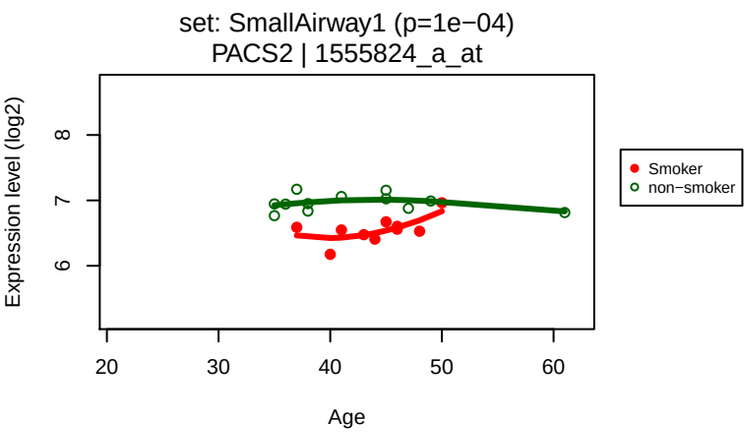
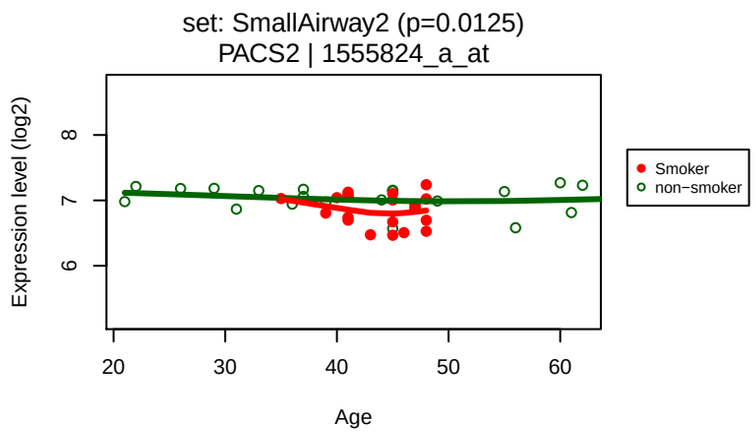
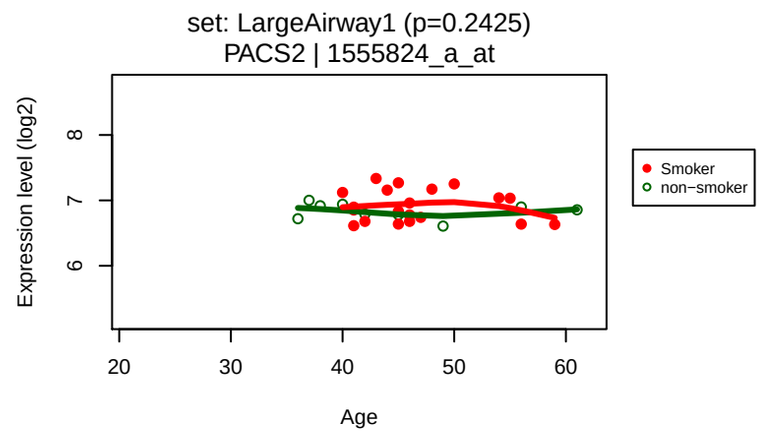

Gene expression of smokers and nonsmokers
gene: PACS2 | 1555824_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.0125
p=0.5649
p=0.2425
p=0.1517
p=0.2654
p=0.9665
p=0.0579
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

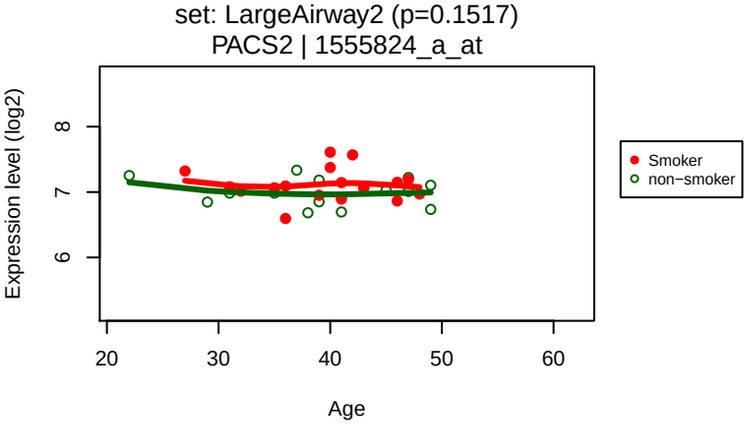

Gene expression ratio (smokers/nonsmokers)
gene: PACS2 | 1555824_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0125
p=0.5649
p=0.2425
p=0.1517
p=0.2654
p=0.9665
p=0.0579
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

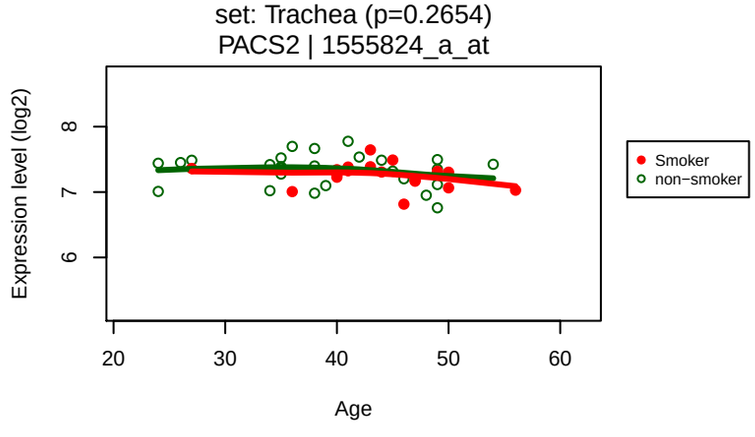

Gene expression ratio in Males
gene: PACS2 | 1555824_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0919
p=0.994
p=0.3301
p=0.0376
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PACS2 | 1555824_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2838
p=0.0273
p=0.8824
p=0.9516
SmallAirway2
LargeAirway2
Trachea
Macrophage2

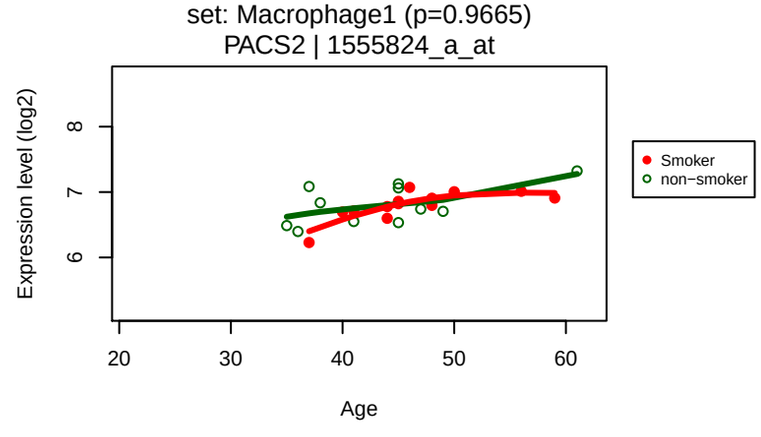

set: SmallAirway1 (p=1e−04)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker

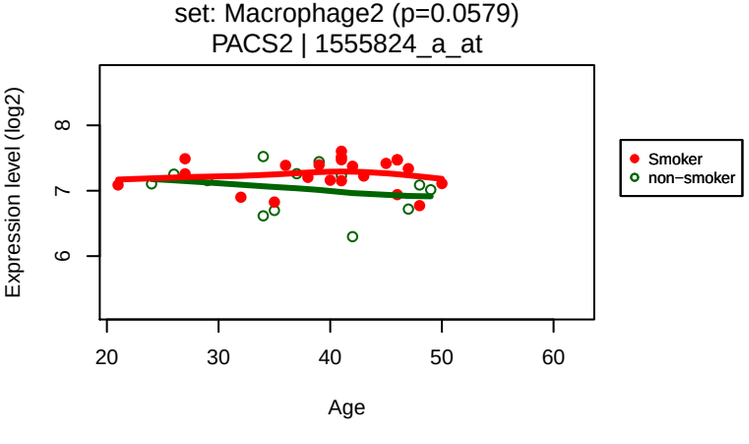

set: SmallAirway2 (p=0.0125)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker

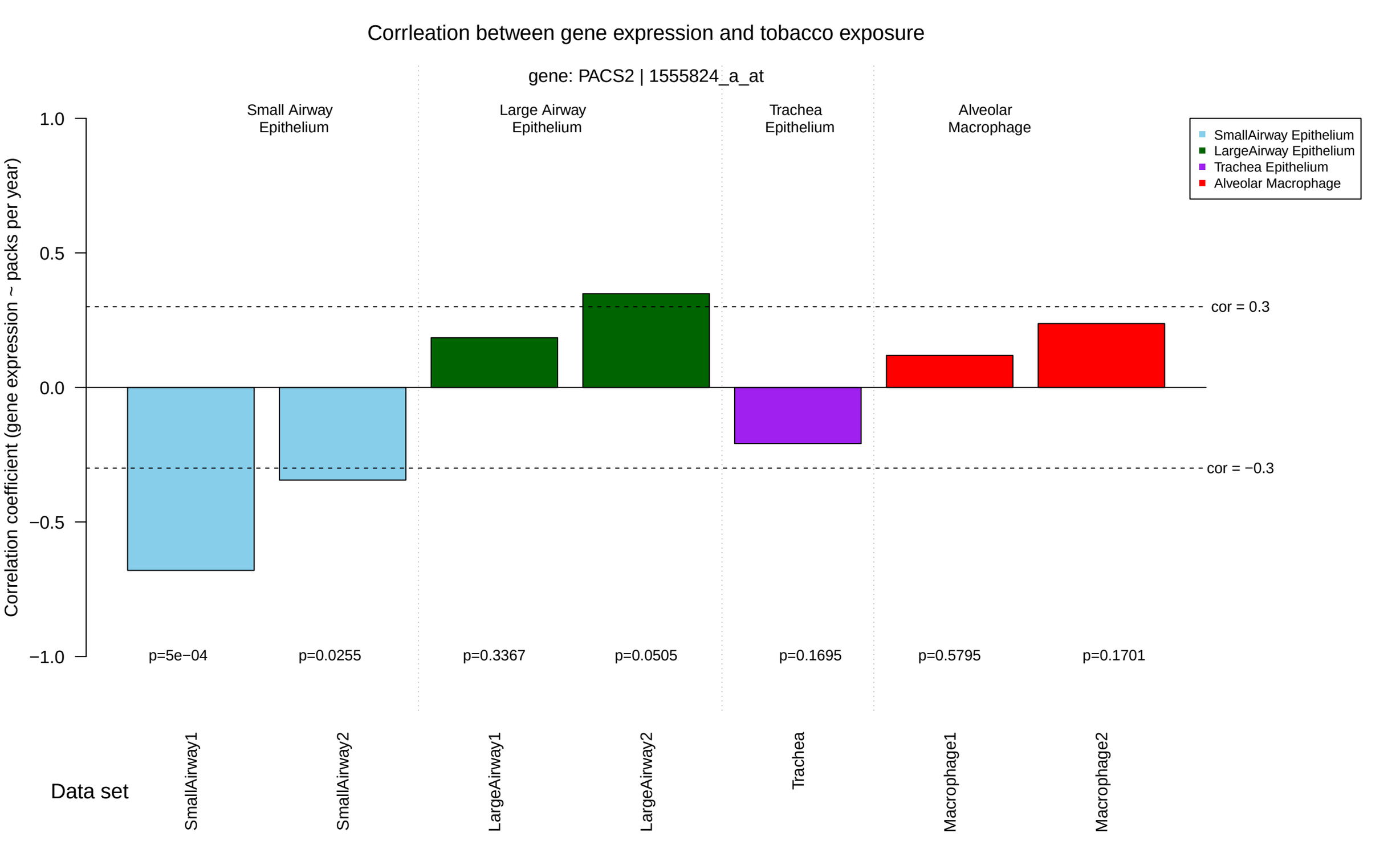

set: LargeAirway1 (p=0.2425)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1517)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2654)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9665)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0579)
PACS2 | 1555824_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PACS2 | 1555824_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=5e−04
p=0.0255
p=0.3367
p=0.0505
p=0.1695
p=0.5795
p=0.1701
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2